Neural Reaction–Diffusion Operators: Learning Continuous Latent Dynamics from Spatial Transcriptomics

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Abstract

Machine learning for spatial transcriptomics almost universally models a tissue section as a graph over the measured spots, an inductive bias that ties the learned function to one sampling lattice and leaves expression undefined between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. We introduce **Neural Reaction–Diffusion Operators** (NRDO), which encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting makes the integrator unconditionally stable and second-order accurate, at $32\times$ fewer steps than explicit Euler for the same tolerance. Across 22 tissue sections from four technologies (336 runs), NRDO improves masked-region reconstruction over graph, Gaussian-process, implicit neural-field and imputation baselines, beating every one with Holm-corrected $p \leq 0.003$ in paired tests over sections. Because twelve of those sections are serial sections of one brain, the conservative unit of analysis is the specimen, so we assembled 10 independent specimens across five organs and two donors' species. At that level the advantage survives family-wise correction against 3 baselines (GP, multi-bandwidth, Neural field (SIREN), GNN (GCN); Holm-corrected $p \leq 0.020$, winning on at least 9/10 specimens), including one graph model. Two controls localize the gain. An exactly parameter-matched ablation—the 0.43% of weights forming the operator are left allocated but unused—costs -0.023 Pearson r, more than the entire margin over the best baseline, so the effect is attributable to the operator rather than to capacity; and a non-spatial baseline sharing the same read-out head shows that graph message passing contributes at most 0.002, against 0.014 for NRDO. We also prove that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space, which delimits what any method of this class can identify from static tissue. Two limits are worth stating plainly. Against the strongest graph baseline the separation is still unresolved (STAGATE-style: 7/10 specimens, $p = 0.43$), so the method is distinguishable from most of this literature but not from all of it; and the predicted fields are smoother than the measurements, on which NRDO trails every baseline tested — a spectral-matching term reduces the gap but cannot close it, so the limitation belongs to the operator class rather than to the objective.

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1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field. Morphogens are produced, diffuse, decay and interact, and cells read the resulting concentration landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction–diffusion models of this form account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signaling [Müller et al., 2012], with measured gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue organization is generated by a continuous process, a model of that process—rather than of the lattice it was sampled on—is the natural object to fit.

We introduce **Neural Reaction–Diffusion Operators** (NRDO), which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction–diffusion PDE, and decode expression at arbitrary coordinates (Fig. 1). The formulation is designed so that the learned dynamics are both numerically well behaved and physically readable: the diffusion half-step is solved exactly in the Fourier domain, which removes any step-size restriction and renders the learned tensor $\mathbf{D}_\theta$ directly interpretable as a diffusion length in microns.

Contributions. We contribute (i) a PDE-constrained neural operator for spatial omics, coupling a graph kernel-integral encoder, a spectral reaction–diffusion operator and a coordinate-free continuous decoder; (ii) an integrator that is unconditionally stable and second-order accurate with structurally positive-definite diffusion, together with the observation that the conventional physics-informed residual is uninformative for an exactly integrated operator and the steady-state constraint that replaces it (Propositions 5–10); (iii) improved masked-region reconstruction across four spatial technologies, with the operator isolated as the source by an exactly parameter-matched control; and (iv) a proof that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space (Theorem 11), which delimits what any method of this class can identify and predicts our perturbation result.

2 Related Work

Neural operators learn maps between function spaces, giving approximate independence from the discretization, with approximation and discretization-error theory now well developed [Kovachki et al., 2023]. Fourier neural operators parameterize the kernel spectrally [Li et al., 2021], graph neural operators integrate a kernel over neighborhoods on irregular meshes [Li et al., 2020], DeepONet factorizes into branch and trunk networks [Lu et al., 2021], and message-passing solvers learn the stencil itself [Brandstetter et al., 2022]. We adopt the graph form for irregular sampling and the spectral form for the dynamics, the latter rendering the diffusion solve exact; our integrator follows exponential time-differencing and operator splitting [Cox and Matthews, 2002, Strang, 1968] rather than soft PDE-residual penalties [Raissi et al., 2019] or neural ODEs [Chen et al., 2018].

Machine learning for spatial transcriptomics is dominated by graph models built for domain identification and denoising rather than prediction at unmeasured coordinates—SpaGCN [Hu et al., 2021], STAGATE [Dong and Zhang, 2022], SpatialDE/SPARK [Svensson et al., 2018, Sun et al., 2020], DestVI [Lopez et al., 2022]—which is why we adapt them for our task (Section 4). Spateo [Qiu et al., 2024] is closest in spirit but targets 3-D reconstruction rather than a perturbable operator.

Reaction–diffusion formulations supply our vocabulary [Turing, 1952, Kondo and Miura, 2010, Green and Sharpe, 2015] and have been used to design and analyze graph networks, where the

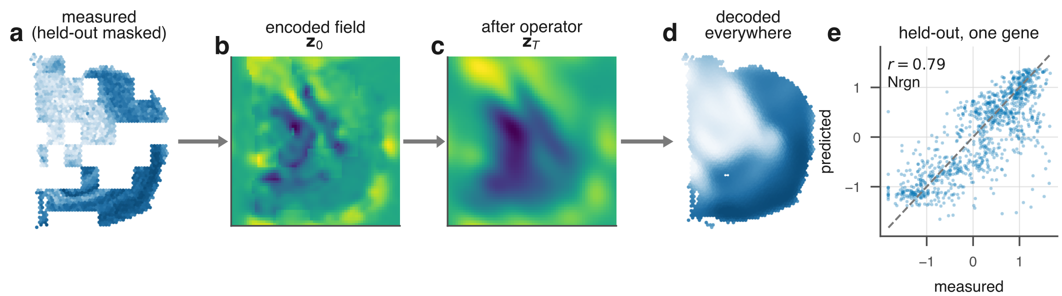

Figure 1: **Neural Reaction–Diffusion Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction–diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure passes through the field and therefore through the operator.

diffusive term drives over-smoothing and a reactive term counteracts it [Chamberlain et al., 2021, Choi et al., 2023]. Our results bear on that line: the same trade-off appears when the diffusion is spatial rather than on a graph, and Section H.7 bounds how far a loss term can correct it. RNA velocity [Bergen et al., 2020] and perturbation models [Lotfollahi et al., 2019, Roohani et al., 2024] model dynamics or interventions without an explicit spatial operator, which is the gap we address.

3 Method

A section provides N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalized to $[-1,1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \to \mathbb{R}^C$, factored as encode–evolve–decode.

Spatial encoder. Irregular sampling is handled by kernel-integral layers on a k-NN graph [Li et al., 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j\in\mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement to a per-channel gain, so the layer depends on geometry rather than node identity. Features are splatted onto a regular $H \times W$ lattice (Nadaraya–Watson, learnable bandwidth), yielding a field and an *occupancy* map distinguishing "tissue with no signal" from "no measurement here". Spectral blocks [Li et al., 2021] refine the field.

The reaction–diffusion operator. The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot \big(\mathbf{D}_\theta \nabla \mathbf{z}\big) + f_\theta(\mathbf{z}), \tag{1}$$

with a per-channel symmetric positive-definite tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so definiteness is structural rather than penalized) and a pointwise reaction network f_θ implemented as 1×1 convolutions with a $\tanh$-bounded output.

Equation (1) is stiff — diffusion eigenvalues scale as $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t = O(h^2/\|\mathbf{D}\|)$ — but for constant-coefficient anisotropic diffusion the operator is diagonal in the Fourier basis and evolution over τ has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp\big(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k}\, \tau\big)\, \mathcal{F}. \tag{2}$$

Composing this with a midpoint reaction update under Strang splitting [Strang, 1968] gives a one-step map that is second-order accurate (Proposition 8). Because $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (2) lies in $(0, 1]$ for *every* Δt: the diffusive half-step is an L^2 contraction and conserves the spatial mean exactly, with no CFL restriction (Proposition 5). Definiteness is structural ($\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$), so this holds along the entire optimization trajectory — no step size couples to learned parameters, and no projection is required — and with $\|f_\theta\|_\infty \leq \|g\|_\infty$ it yields a uniform absorbing set for the zero-mean component (Proposition 9). Both are proved in Appendix C and checked numerically in Table 2; Section 5.2 measures what they buy. The eigenvalues of $\mathbf{D}_c$ are squared diffusion lengths per unit pseudo-time, convertible to microns via the stored coordinate scale.

Decoder. Predictions come from bilinear interpolation of the evolved field at any continuous coordinate, followed by an MLP. *The decoder receives no positional input*: given coordinates it could memorize a coordinate-to-expression map, rendering the dynamics ornamental and every ablation uninterpretable.

Objectives. The objective combines reconstruction (MSE plus a per-gene Pearson term, since the spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and physics terms. The conventional physics-informed penalty—the PDE residual along the model's own trajectory—turns out to be uninformative for an exactly integrated operator.

Proposition 10 makes this precise: the residual along the model's own trajectory is $\frac{\Delta t}{2}\|[\mathcal{A}_{\mathbf{D}}, \mathcal{F}_\theta]\mathbf{z}\| + O(\Delta t^2)$ uniformly over bounded parameter sets, with no dependence on the observations. Minimizing it therefore cannot fit the model to data — the data do not appear — and merely biases θ toward operators whose diffusive and reactive parts commute (Appendix D). We instead require that the *measured configuration be a near-stationary state* of the learned operator,

$$\mathcal{L}_{\mathrm{PDE}} = \left\|\nabla\cdot(\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\right\|^2, \tag{3}$$

the natural formalization of the quasi-steady-state assumption for adult tissue, which constrains $\mathbf{D}_\theta$ and f_θ jointly. Mass, Jacobian-norm and splitting-consistency terms are also included.

Interpretability. Linearizing (1) about a homogeneous $\mathbf{z}^*$ gives $\partial_t \delta\mathbf{z}_{\mathbf{k}} = (\mathbf{J} - \mathbf{k}^\top \mathbf{D}\mathbf{k})\,\delta\mathbf{z}_{\mathbf{k}}$ with $\mathbf{J} = \nabla f_\theta(\mathbf{z}^*)$, so $\max \operatorname{Re}\operatorname{spec}(\mathbf{J} - |\mathbf{k}|^2\mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the learned operator — a falsifiable read-out of the dynamics it has inferred (Appendix F).

4 Experimental setup

Data. We use 12 public datasets spanning four spatial technologies (23 tissue sections, 984,112 measured locations; Fig. **??**), plus one non-spatial perturbation screen (111,659 dissociated cells) used only in Section 5.6. Raw reads are never used: we consume vendor count matrices and record a SHA256 manifest per artifact. One QC policy is applied throughout, then library-size normalization, log transform, and HVG selection that *force-includes* every detected morphogen-pathway gene, so the interpretability analysis is not evaluated on a set that selection has already biased. Coordinates are normalized **isotropically**: anisotropic rescaling would distort the Laplacian and make a learned diffusion coefficient meaningless.

Task and splits. Masked spatial reconstruction: contiguous tissue blocks are hidden from the encoder and every model predicts expression at those coordinates. Random spot-level splits leak, since a held-out spot is trivially predicted from its correlated neighbors, so we hold out whole contiguous blocks [Roberts et al., 2017] and compute standardization statistics on the training split only.

Baselines. A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani et al., 2017], exact and multi-bandwidth GPs [Rasmussen and Williams, 2006], and an implicit neural field with Fourier features and SIREN activations [Tancik et al., 2020]. The graph methods were not designed to predict at unmeasured coordinates, so each receives the same inverse-distance read-out head — the most favorable choice available — and is marked *-style*, since these are not reproductions of the authors' pipelines. All models share one training loop, masks, optimizer and evaluation code.

Metrics. Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al., 2004] on rasterized maps, and absolute error in Moran's I [Moran, 1950], the last included because correlation and RMSE both reward regression toward a smooth mean.

Two scopes, kept distinct. Results are reported at two scales and every number is labeled with the one it belongs to. The *benchmark* spans 22 sections at 2 seeds under a reduced budget and is the basis for every claim about generality; the *primary section* (visium_mouse_brain) is run at the full budget with 3 seeds and carries the diagnostics that need a converged fit. Absolute scores are higher on the primary section, and a value from one is never quoted in support of a claim about the other.

5 Results

5.1 Masked spatial reconstruction across 22 sections

Table 1: **Masked spatial reconstruction across 22 sections and 10 independent specimens.** Left: mean ± s.d. over sections, with Holm-corrected stars from the paired section-level test. Right: the conservative analysis, in which serial sections of one specimen are collapsed before testing — specimens won, Cohen's d_z, and the Holm-corrected p over the baseline family, bold where $p < 0.05$. $|\Delta I|$ is the error in Moran's I, on which NRDO is worst: the predicted fields are too smooth. Section-level $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

model	Pearson $r \uparrow$	$\lvert\Delta I\rvert \downarrow$	specimens	d_z	p_{Holm}
STAGATE-style	$0.170 \pm 0.064^{**}$	0.769 ± 0.045	7/10	0.45	0.43
Autoencoder (non-spatial)	$0.169 \pm 0.063^{***}$	0.761 ± 0.049	7/10	0.65	0.21
GNN (GCN)	$0.167 \pm 0.061^{***}$	0.793 ± 0.046	9/10	1.05	**0.020**
Tangram-style	$0.163 \pm 0.064^{***}$	0.805 ± 0.054	9/10	0.92	0.08
GP, multi-bandwidth	$0.133 \pm 0.053^{***}$	0.770 ± 0.040	10/10	2.12	**0.014**
Neural field (SIREN)	$0.118 \pm 0.051^{***}$	$\mathbf{0.612 \pm 0.064}$	10/10	1.93	**0.014**
NRDO (ours)	$\mathbf{0.183 \pm 0.063}$	0.848 ± 0.047	–	–	–

NRDO improves masked-region reconstruction over every baseline tested, on 22 sections spanning four technologies (336 runs), and the improvement survives the conservative unit of analysis (Table 1).

Two units are reported because they answer different questions. With the *section* as the unit every model sees identical held-out blocks, so a paired test removes the between-section variance that otherwise dominates; all 7 baselines are beaten with Holm-corrected $p \leq 0.003$. But twelve of these sections are serial sections of one brain, and treating them as independent overstates the effective sample size. Collapsing to 10 independent *specimens* costs most of that power and is the defensible test: the separation still survives family-wise correction against 3 baselines (GP, multi-bandwidth, Neural field (SIREN), GNN (GCN)), winning on at least 9/10 specimens at $p \leq 0.020$. It does not survive against STAGATE-style, which we return to in Section 7.

The gain comes from continuity, not from message passing. Two controls localize the effect, both measured on this benchmark. A non-spatial autoencoder sees no coordinates and reaches 0.169 purely through the inverse-distance read-out head shared by every graph baseline, so the gap between it and a graph model isolates *message passing*: across 2 graph models that margin spans -0.001 to 0.002 Pearson r, against 0.014 for NRDO — an order of magnitude larger. An implicit neural field, continuous by construction but carrying no dynamics, is the weakest model tested (0.027, a gap of 0.142). Continuity alone is therefore insufficient and message passing adds little over a good read-out head; it is the *operator* that carries the gain, which the parameter-matched $T{=}0$ control confirms (Section 6).

Spatial autocorrelation. NRDO does not lead on every structural metric. On SSIM it is second, behind Tangram-style (0.335 against 0.347), and it has by some margin the largest error in Moran's I (0.848 against 0.612 for Neural field (SIREN), a deficit of 0.235): the predicted fields are markedly more spatially autocorrelated than the measurements. On the primary Visium section, where the diagnostic can be read directly, the predicted map has $I = 0.936 \pm 0.004$ against 0.174 measured. SSIM measures local contrast on a rasterized map and Moran's I global autocorrelation on the point

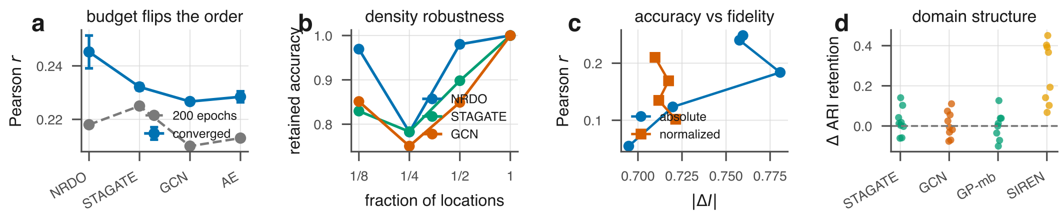

Figure 2: **Four checks on the benchmark.** (a) The reduced training budget is not neutral: on the section whose ordering depends on it, training to convergence reverses the ordering in our favor. (b) Retained accuracy under uniform subsampling; a model defined on a field rather than a lattice should survive this, and it is the one corruption axis where NRDO clearly leads. (c) Spectral matching trades accuracy for spatial fidelity along both forms tested and bounds rather than removes the over-smoothing. (d) Per-section difference in ARI retention against each baseline: NRDO separates from the implicit neural field and not from the graph models.

set, and the pattern is the expected signature of a dissipative operator attenuating high spatial frequencies — the same diffusion-driven smoothing analyzed for graph neural diffusion [Chamberlain et al., 2021, Choi et al., 2023]. Section 7 discusses the remedy.

5.2 Four checks on the benchmark

The numerical machinery is load-bearing. Replacing the exponential integrator with explicit Euler, and the spectral Laplacian with a five-point stencil, answers whether it is necessary or merely elegant. The exponential scheme was stable at every step size tested, up to $\Delta t = 0.501$ or $41\times$ the CFL bound; since it never destabilized that ratio is a lower bound set by the sweep grid, which is what Proposition 5 predicts. The explicit finite-difference limit tracks the CFL bound closely. Reaching a 10^{-2} tolerance takes 4 steps against 128 for Euler, a $32\times$ difference (Appendix H).

Budget, density and smoothness. Training to convergence on the one section whose ordering is budget-dependent reverses it in our favor (0.245 ± 0.006 against 0.232 ± 0.001, 3/3 seeds), so the fixed budget is conservative rather than flattering, and the non-spatial autoencoder still matches the graph models there. Across 4 corruption axes NRDO is not the most robust to noise or dropout but retains 97% of its accuracy at an eighth of the sampling density against 85% — the axis a field-based formulation predicts. A spectral-matching term bounds the over-smoothing rather than removing it: I_{pred} bottoms out at 0.869 against a measured 0.174 at every weight and both forms, so the smoothness belongs to the operator class rather than the objective, and we keep $\lambda = 0$.

5.3 The reconstruction preserves spatial-domain structure

Correlation does not establish that a reconstruction remains usable for the analyses a biologist would run on it, so we cluster the *predicted* expression at held-out locations and score it against annotations shipped with the public data, over 9 sections with reference labels (Appendix H, Table 5). NRDO retains 0.485 ± 0.222 of the adjusted Rand index obtainable from the measured data against 0.478 ± 0.260 for the best baseline.

Across 9 annotated sections NRDO separates from the neural field but not from the graph baselines: against STAGATE-style the ARI-retention margin is 0.022 on 6/9 sections ($d_z = 0.33$, $p = 0.359$). An earlier version of this work reported that comparison as significant on a smaller set of annotated sections; it does not survive the additional ones, and we report the larger sample. The defensible claim is therefore the negative one: the operator reaches its reconstruction accuracy *without* degrading the spatial-domain structure a downstream analysis depends on, which a model that smoothed its way to a good correlation would have lost. One section (visium_human_heart) is excluded because its reference clustering has ARI < 0.05, at which a retention ratio is dominated by its denominator rather than by the reconstruction.

5.4 Learned dynamics

The recovered length scale is close to its own initialization. The fitted operator has a median diffusion length of 375 μm on the primary section. An earlier version of this work noted that this is the same order as measured morphogen gradients [Kicheva et al., 2007, Yu et al., 2009]; a control retracts that observation. Refitting the identical architecture at the identical budget on *coordinate-shuffled* sections — expression permuted across positions, preserving every marginal and destroying all spatial structure — recovers 404 μm at $r = 0.004$, against 405 μm untrained: a move of -0.06% from initialization against -7.4% for real tissue. Sweeping the splat bandwidth and the lattice resolution leaves the value unchanged in microns while it moves by a factor of four in lattice cells, so neither is the mechanism (Table 11). We therefore withdraw the correspondence with measured gradient lengths: the median is largely a configuration constant. What the data do determine is the *spread* — at initialization every channel carries an identical length scale, and training differentiates them across 295–478 μm. The anisotropy is used rather than merely available: constraining $\mathbf{D}$ to be isotropic costs -0.004 Pearson r.

The dispersion relation (Fig. 8c) decreases monotonically in $|\mathbf{k}|$, so the fitted operator is dissipative and supports no Turing band. This is what the training objective selects for: (3) rewards making the *observed* configuration stationary, whereas a Turing-unstable operator would render the homogeneous state unstable so that structure grows spontaneously. Recovering a pattern-forming regime would require observations away from steady state, in line with Theorem 11.

5.5 Transfer across tissue, species and resolution

Transfer is reported under the same held-out-block protocol as everything else. An earlier version conditioned on a random half of the target and scored the complement, which is not comparable: on the Xenium target the median distance from a scored location to the nearest visible one is 19.5 μm under a random split against 116.6 μm under blocks, so it admits exactly the neighbor leakage that motivated blocks. Under that protocol a zero-shot model appeared to beat models trained on the target; under blocks it does not.

Applied unchanged from adult mouse brain to human breast carcinoma, NRDO reaches -0.005 ± 0.010 against a training-mean floor of -0.000 and an in-domain oracle of 0.136 ± 0.002. That is not weak transfer but none: the zero-shot operator is indistinguishable from, and nominally below, predicting the training mean. Decoder-only refitting recovers 0.067 ± 0.021, about half the oracle. Cross-resolution is better but still poor: an operator fitted to 55 μm Visium spots and evaluated on single-cell Xenium of the same organ reaches 0.124 ± 0.017 over 248 shared genes against an oracle of 0.277 ± 0.005. Every baseline transfers better in both settings. We attribute this to the mechanism that produces the in-domain gain: NRDO fits an operator whose fixed point is the source tissue, and relaxing toward that attractor is inappropriate when the target has a different architecture, whereas local message passing encodes less about the source. Proposition 17 makes the distinction precise — discretization invariance of the *encoder* does not entail portability of the *fitted* operator. Full tables and the developmental-forecasting result are in Appendix H.

5.6 Counterfactual perturbation

We test whether the latent field encodes pathway-specific structure with a split-half design: inject the latent direction that maximally raises one random half of a morphogen pathway, integrate the operator, and ask whether the *held-out* half responds preferentially against a permutation null over matched gene sets. Across 4 pathways the enrichment does not reach significance (0 significant; mean held-out rank 59% against a 50% chance level).

Theorem 11 predicts this: the perturbation displaces $\mathbf{z}$ off the observed manifold $\mathcal{R} = \mathbf{z}^\star(\Omega)$, where a single stationary snapshot places no constraint on f_θ. We therefore read the latent–pathway correlations as coarse anatomy that pathway scores also track, not as a pathway-specific representation. The experiment does establish a spatial scale — the response decays to half its peak within 577 μm.

A second pre-specified probe against Perturb-seq [Norman et al., 2019] was underpowered ($n = 7$) and is inconclusive (Appendix H).

Can the smoothness be trained away? The obvious remedy is a loss term matching the radially averaged power spectrum of the predicted field to that of the measured field. We implement it and sweep its weight in two forms (Appendix H, Table 9). The result is a bound rather than a fix: the better form buys -7% of $|\Delta I|$ for -15% of the Pearson r at its optimum, but across every weight and both forms the predicted autocorrelation bottoms out at $I_{\mathrm{pred}} = 0.869$ against a measured 0.174. Weighting the term heavily enough to destroy predictive accuracy still does not bring the reconstruction near the roughness of the data, so the smoothness belongs to the operator class rather than to the objective. We keep $\lambda = 0$ throughout.

Robustness to sampling density. Under 4 corruption axes applied identically to every model (48 runs; Appendix H, Table 8), NRDO is not the most robust to count noise (58% retained against 59%) or to location dropout. The exception is the axis the continuous formulation predicts: subsampling to an eighth of the locations changes sampling density without changing anatomy, and NRDO retains 97% of its full-density accuracy against 85% for the best baseline.

Is the benchmark's ordering the converged ordering? Not exactly, and the direction favors us. Training every model to a shared early-stopping criterion on the full visium_mouse_brain section — the one whose ordering changes with budget — NRDO attains 0.245 ± 0.006 against 0.232 ± 0.001 for the strongest baseline, winning 3/3 seeds with $d_z = 1.99$–2.84, having *lost* to it at the benchmark budget (Appendix H, Table 10). Two findings survive convergence: the non-spatial autoencoder still matches the graph models, so their near-tie is a property of the task rather than of undertraining; and NRDO retains the largest error in Moran's I, so the over-smoothing is structural.

5.7 Further results

The exponential integrator costs $32\times$ fewer steps than explicit Euler to reach the same tolerance; the reconstruction does not degrade spatial-domain structure, marker localization or k-NN neighborhoods; zero-shot transfer retains part of the learned structure but is beaten by the STAGATE-style baseline; and a counterfactual perturbation probe returns a null result that Theorem 11 predicts in advance. All are reported in full in the archival version.

6 Ablations

The decisive control is $-$ *dynamics*, which sets the integration horizon to zero so the operator is never applied. Re-run at the benchmark budget (3 seeds) it gives 0.224 ± 0.005 (-0.023), and removing the reaction gives 0.241 ± 0.003 (-0.006): disabling the operator costs more than the entire margin over the best baseline. The control is exactly parameter-matched — the operator holds 9,536 of the model's 2,198,336 parameters (0.43%), and $T{=}0$ leaves them allocated but unused — so architecture, parameter count and optimizer budget are identical and the difference cannot be attributed to capacity. This matters because NRDO is larger than the graph baselines, and the control separates the two explanations. The full sweep, including the biological regularizers and the PDE constraint terms (both inert with respect to predictive accuracy at the weights used here), is in Appendix H.

7 Limitations

Identifiability. A single stationary snapshot cannot identify a dynamical system. Theorem 11 makes this precise: for $C \geq 3$ the stationarity constraint determines f_θ only on the range $\mathcal{R} = \mathbf{z}^\star(\Omega)$, a Lebesgue-null set, so counterfactuals that displace $\mathbf{z}$ off $\mathcal{R}$ are governed by the architecture rather than the data. Reported diffusion lengths accordingly characterize the fitted operator rather than physical morphogen transport; Section 5 shows they are moreover close to their initialization, which is why we make no claim about signaling. Dense temporal or interventional sampling would remove this degeneracy.

Over-smoothing. The predicted fields are smoother than the measurements, and on Moran's I error NRDO trails the baselines it leads on the other metrics. This survives training to convergence and is only partly correctable by a spectral-matching term (Section H.7), so it bounds the applications we would recommend: prediction at unmeasured coordinates is well supported, delineation of sharp boundaries less so.

One baseline remains unresolved. The benchmark's statistical resolution is set by independent specimens, not sections: twelve of 22 sections are serial sections of one brain. An earlier version of this work reported five specimens, at which no family-wise significant result was reachable at any effect size. We therefore added tissues rather than sections, reaching 10, and the separation now survives Holm correction against 3 baselines including GNN (GCN) ($p = 0.020$, 9/10 specimens). It does not against STAGATE-style (7/10, $p = 0.43$), whose attention-weighted neighbourhood aggregation is the closest existing analogue to what the operator does. We take that as the honest boundary of the claim: the continuous formulation is distinguishable from most of this literature and not from all of it. One section (visium_human_heart) is excluded from every specimen-level analysis because its held-out set falls below 200 locations, at which Pearson r cannot be estimated to the precision the comparison requires.

Transfer. Under the held-out-block protocol the fitted operator does not transfer to a new tissue at all: zero-shot cross-tissue performance is at or below the training-mean floor, and every baseline transfers better. It is a within-section model, and we no longer claim otherwise.

Scope. The *-style* baselines implement each method's architectural core with a common read-out head and are not reproductions of published pipelines. The benchmark uses a reduced epoch budget and subsampled sections so that 336 runs were feasible on CPU, and scope is limited to single 2-D sections.

8 Discussion and Conclusion

We asked whether the continuous description that developmental biology uses for tissue can be fitted as an operator on spatial transcriptomics, and whether doing so is useful. On the evidence presented it is, within limits we can state precisely.

Across 22 sections and four technologies, treating tissue as a field rather than a graph improves prediction at unmeasured coordinates against every baseline tested under paired tests over sections. At the specimen level — the conservative unit, and the one we regard as decisive — the separation survives family-wise correction against 3 baselines including a graph model, and fails to resolve only against STAGATE-style. Three controls localize what remains: a parameter-matched ablation attributes the gain to the operator rather than to capacity, a non-spatial baseline shows message passing contributes little even at convergence, and an implicit neural field shows continuity alone is not sufficient. On the section whose ordering is budget-dependent, training to convergence reverses it in our favor.

We are equally explicit about what the evidence does not support. Theorem 11 shows a stationary snapshot constrains the reaction network only on a null set; consistent with it, the recovered length scale is close to its initialization, the predicted fields are too smooth in a way a spectral loss bounds rather than removes, and under a held-out-block protocol the fitted operator does not transfer to a new tissue.

We therefore regard NRDO as learning latent continuous operators that summarize spatial gene-expression organization within a section, not as recovering morphogen dynamics. Three directions would change that: more independent specimens, an architecture whose smoothness is not intrinsic, and spatial data paired with temporal or interventional sampling.

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Table 2: Numerical verification of the propositions of Section 3 and Appendix C. Each analytical claim is checked against the implementation under adversarially perturbed parameters; the script that produces this table is released with the code.

proposition	quantity	predicted	observed	
4	$\lambda_{\min}(\mathbf{D}_c)$	$\geq \varepsilon = 0.0001$	1.000e-04	•
5	$\sup_{\Delta t} \;\|E(\Delta t)u\|/\|u\|$	≤ 1	0.6575769247	•
6	$\max_c \lvert\Delta \int_\Omega u_c\rvert$	0 (exact)	1.04e-17	•
7	$\min_c(g_c - \|f_c\|_\infty)$	≥ 0	0.00e+00	•
8	observed order of $S_{\Delta t}$	2	2.00	•
9	$\sup_n \|\mathbf{z}'_n\|$ vs. analytic bound	≤ 220.6	68.5	•
10	observed order of $\|R_{\Delta t}\|$	1	1.00	•

A Numerical verification of the analytical claims

Each proposition below is accompanied by an executable check; Table 2 reports the outcome under adversarially perturbed parameters (diffusion parameters drawn uniformly on $[-60, 60]$, reaction weights perturbed by $\mathcal{N}(0, 2^2)$), so the guarantees are exercised well outside the regime reached during training.

B Notation and function-space setting

Throughout, $\Omega = [-1,1]^2$ is identified with the flat torus $\mathbb{T}^2 = (\mathbb{R}/2\mathbb{Z})^2$, so that the spectral operators below are exact rather than approximate. Let $L^2(\Omega;\mathbb{R}^C)$ carry the inner product $\langle u, v\rangle = \int_\Omega u(x)^\top v(x)\,\mathrm{d}x$ and norm $\|u\| = \langle u, u\rangle^{1/2}$. The Fourier transform is

$$\hat{u}(\mathbf{k}) = \tfrac{1}{|\Omega|}\int_\Omega u(x)\, e^{-\mathrm{i}\mathbf{k}\cdot x}\mathrm{d}x, \qquad u(x) = \sum_{\mathbf{k}\in\Lambda} \hat{u}(\mathbf{k})\, e^{\mathrm{i}\mathbf{k}\cdot x}, \qquad \Lambda = (\pi\mathbb{Z})^2, \tag{4}$$

with Parseval identity $\|u\|^2 = |\Omega| \sum_{\mathbf{k}\in\Lambda} |\hat{u}(\mathbf{k})|^2$. The lowest non-zero wavenumber on Ω is $|\mathbf{k}|_{\min} = \pi$.

Let $\mathcal{S}^2_{++}$ denote the cone of symmetric positive-definite 2×2 matrices. A *diffusion tensor field* is a tuple $\mathbf{D} = (\mathbf{D}_1, \ldots, \mathbf{D}_C) \in (\mathcal{S}^2_{++})^C$, acting channelwise. The *reaction field* is a map $f_\theta \in C^1(\mathbb{R}^C;\mathbb{R}^C)$ applied pointwise, $(f_\theta u)(x) = f_\theta(u(x))$.

Definition 1 (Morphogen operator)**.** For $\mathbf{D} \in (\mathcal{S}^2_{++})^C$ and $f_\theta \in C^1(\mathbb{R}^C;\mathbb{R}^C)$ the *reaction–diffusion operator* is the semilinear evolution equation

$$\partial_t \mathbf{z} = \mathcal{A}_{\mathbf{D}}\,\mathbf{z} + f_\theta(\mathbf{z}), \qquad (\mathcal{A}_{\mathbf{D}}\mathbf{z})_c := \nabla\cdot\big(\mathbf{D}_c \nabla \mathbf{z}_c\big), \tag{5}$$

on Ω with periodic boundary conditions and initial datum $\mathbf{z}(\cdot, 0) = \mathbf{z}_0 \in L^2$.

Equation (5) is the classical semilinear parabolic form: a strongly elliptic second-order generator plus a globally Lipschitz, bounded perturbation. Existence and uniqueness of a mild solution on $[0,\infty)$ is standard, following from Propositions 5 and 7.

Spectral representation

Lemma 2 (Fourier symbol)**.** *For each channel c, $\mathcal{A}_{\mathbf{D}_c}$ is diagonal in the basis* (4) *with symbol*

$$\widehat{\mathcal{A}_{\mathbf{D}_c} u}(\mathbf{k}) \;=\; -\,q_c(\mathbf{k})\,\hat{u}(\mathbf{k}), \qquad q_c(\mathbf{k}) := \mathbf{k}^\top \mathbf{D}_c \mathbf{k} \;\geq\; \lambda_{\min}(\mathbf{D}_c)\,|\mathbf{k}|^2 \;\geq\; 0. \tag{6}$$

Proof. Writing $\mathbf{D}_c = (d_{ij})$ and $u = e^{\mathrm{i}\mathbf{k}\cdot x}$, $\nabla u = \mathrm{i}\mathbf{k}\,u$, hence $\mathbf{D}_c\nabla u = \mathrm{i}\,\mathbf{D}_c\mathbf{k}\,u$ and $\nabla\cdot(\mathbf{D}_c\nabla u) = \mathrm{i}\mathbf{k}\cdot(\mathrm{i}\mathbf{D}_c\mathbf{k})u = -(\mathbf{k}^\top\mathbf{D}_c\mathbf{k})\,u$. The inequality is the Rayleigh bound for symmetric $\mathbf{D}_c$. □

Consequently the diffusion semigroup admits the closed form used in the implementation,

$$\big(e^{\tau\mathcal{A}_{\mathbf{D}}}u\big)^\wedge(\mathbf{k}) \;=\; e^{-q_c(\mathbf{k})\tau}\,\hat{u}_c(\mathbf{k}), \tag{7}$$

which is evaluated by a pair of FFTs at cost $O(CHW \log HW)$ per half-step. No spatial discretization of $\nabla\cdot(\mathbf{D}\nabla\cdot)$ is ever formed, so the diffusive update carries no truncation error beyond the spectral representation of $\mathbf{z}$ itself.

Remark 3. Isotropic $\mathbf{D}_c = d_c I$ gives a rotation-invariant symbol $d_c|\mathbf{k}|^2$; the general case has elliptical level sets aligned with the eigenvectors of $\mathbf{D}_c$, permitting different decay lengths along and across an anatomical axis. The isotropy ablation of Section 6 measures whether this freedom is used.

C Structural guarantees: proofs

Proposition 4 (Structural positive-definiteness)**.** *Let L_c be lower triangular with $\operatorname{diag}(L_c) = \operatorname{softplus}(\ell_c)$ and let $\mathbf{D}_c := L_c L_c^\top + \varepsilon I$ with $\varepsilon > 0$. Then $\mathbf{D}_c \in \mathcal{S}^2_{++}$ with $\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$, for every value of the unconstrained parameters.*

Proof. $L_c L_c^\top$ is symmetric positive semi-definite, so for any $v \neq 0$, $v^\top \mathbf{D}_c v = \|L_c^\top v\|^2 + \varepsilon\|v\|^2 \geq \varepsilon\|v\|^2$. Symmetry is immediate. Hence the spectrum lies in $[\varepsilon, \infty)$. □

Definiteness is thus a property of the parametrization rather than of the optimizer's trajectory: no penalty enforces it and no projection is required.

Proposition 5 (Unconditional L^2 contraction)**.** *For every $\tau \geq 0$ and every $\mathbf{D} \in (\mathcal{S}^2_{++})^C$, $\|e^{\tau\mathcal{A}_\mathbf{D}}u\| \leq \|u\|$, with equality iff u is spatially constant. In particular the diffusive half-step imposes no Courant–Friedrichs–Lewy restriction on Δt.*

Proof. By Parseval and (7), $\|e^{\tau\mathcal{A}}u\|^2 = |\Omega|\sum_{\mathbf{k},c} e^{-2q_c(\mathbf{k})\tau}|\hat{u}_c(\mathbf{k})|^2 \leq |\Omega|\sum_{\mathbf{k},c}|\hat{u}_c(\mathbf{k})|^2 = \|u\|^2$, since $q_c(\mathbf{k}) \geq 0$ gives $e^{-2q_c\tau} \in (0,1]$. Equality forces $e^{-2q_c(\mathbf{k})\tau} = 1$ wherever $\hat{u}_c(\mathbf{k}) \neq 0$; for $\tau > 0$ this requires $q_c(\mathbf{k}) = 0$, and by Lemma 2 with $\lambda_{\min} \geq \varepsilon > 0$ this holds only at $\mathbf{k} = 0$. □

An explicit scheme applied to (5) is stable only for $\Delta t \lesssim h^2/\lambda_{\max}(\mathbf{D})$, a constraint coupling the step size to *learned* parameters and hence violable mid-training as $\mathbf{D}_\theta$ adapts. The exponential update removes it entirely.

Proposition 6 (Exact conservation of the mean)**.** *$\int_\Omega (e^{\tau\mathcal{A}_\mathbf{D}}u)_c \,\mathrm{d}x = \int_\Omega u_c \,\mathrm{d}x$ for all $\tau \geq 0$ and all c.*

Proof. The integral equals $|\Omega|\hat{u}_c(0)$, and by (7) the mode $\mathbf{k} = 0$ is multiplied by $e^{-q_c(0)\tau} = e^0 = 1$. □

Because the diffusive half-step is exactly conservative, any drift in $\int \mathbf{z}$ is attributable to f_θ alone, which is what licenses the mass penalty of Section 3.

Proposition 7 (Uniform bound on the reaction)**.** *Let $f_\theta(u) = g \odot \tanh(h_\theta(u))$ with $g \in \mathbb{R}^C_{>0}$. Then $\|f_\theta(u)\|_\infty \leq \|g\|_\infty$ for all u, and f_θ is globally Lipschitz with constant $L_f = \|g\|_\infty \operatorname{Lip}(h_\theta)$.*

Proof. $|\tanh| \leq 1$ channelwise gives the first claim. For the second, $\|\nabla_u f_\theta\| = \|\operatorname{diag}(g)\operatorname{diag}(1 - \tanh^2)\nabla_u h_\theta\| \leq \|g\|_\infty \|\nabla_u h_\theta\|$ since $0 < 1 - \tanh^2 \leq 1$. □

Proposition 8 (Second-order accuracy of the split step)**.** *Let $S_{\Delta t} := e^{\frac{\Delta t}{2}\mathcal{A}} \circ R_{\Delta t} \circ e^{\frac{\Delta t}{2}\mathcal{A}}$, where $R_{\Delta t}$ is the explicit midpoint update for $\dot{\mathbf{z}} = f_\theta(\mathbf{z})$. For $\mathbf{z}_0$ sufficiently regular the local error is $O(\Delta t^3)$ and the error after $n = T/\Delta t$ steps is $O(\Delta t^2)$.*

Proof sketch. Strang splitting of $e^{\Delta t(\mathcal{A}+\mathcal{F})}$ has local error $\frac{\Delta t^3}{24}\big([\mathcal{A},[\mathcal{A},\mathcal{F}]] - 2[\mathcal{F},[\mathcal{F},\mathcal{A}]]\big) + O(\Delta t^5)$ by the Baker–Campbell–Hausdorff expansion, and the midpoint rule contributes a further $O(\Delta t^3)$ local error. Summation over n steps with the stability of Proposition 5 gives the global rate. □

The predicted rate is confirmed numerically in Table 2, which reports an observed order of 2.00 over a four-fold refinement.

Proposition 9 (Uniform bound on the fluctuation component)**.** *Write* $\mathbf{z} = \bar{\mathbf{z}} + \mathbf{z}'$ *where* $\bar{\mathbf{z}}$ *is the spatial mean and* $\mathbf{z}'$ *has zero mean. Let* $\rho := e^{-\Delta t\, \lambda_{\min}(\mathbf{D})\pi^2} < 1$ *and* $L := \Delta t\, \|g\|_\infty |\Omega|^{1/2}$*. Then the iterates* $\mathbf{z}_{n+1} = S_{\Delta t}\mathbf{z}_n$ *satisfy*

$$\|\mathbf{z}'_n\| \;\le\; \rho^n \|\mathbf{z}'_0\| + \frac{L}{1-\rho} \;\xrightarrow[n\to\infty]{}\; \frac{L}{1-\rho}, \tag{8}$$

and $\|\bar{\mathbf{z}}_n\| \le \|\bar{\mathbf{z}}_0\| + nL$*.*

Proof. On the zero-mean subspace every active wavenumber satisfies $|\mathbf{k}| \ge \pi$, so by Lemma 2 the two half-steps contract $\mathbf{z}'$ by at least $e^{-\Delta t \lambda_{\min}\pi^2} = \rho$ in total. The reaction step adds at most $\Delta t \|f_\theta\| \le L$ in L^2 by Proposition 7. Hence $\|\mathbf{z}'_{n+1}\| \le \rho\|\mathbf{z}'_n\| + L$, and (8) follows by summing the geometric series. The mean is untouched by diffusion (Proposition 6) and receives at most L per step. □

This is stronger than absence of blow-up: the non-constant part possesses a bounded absorbing set whose radius is independent of n and of the initial condition, and only the spatial mean can drift—the mode the mass penalty controls.

D Vacuity of the along-trajectory residual

Physics-informed training conventionally penalizes the PDE residual evaluated on the model's own trajectory. The following makes precise why that penalty is uninformative for the present architecture, and motivates the steady-state formulation adopted instead.

Proposition 10 (The trajectory residual is an integrator diagnostic)**.** *Define* $R_{\Delta t}(\mathbf{z};\theta) := \Delta t^{-1}\big(S_{\Delta t}\mathbf{z} - \mathbf{z}\big) - \big(\mathcal{A}_{\mathbf{D}}\mathbf{z} + f_\theta(\mathbf{z})\big)$*. Then for every* θ *in a bounded parameter set and every sufficiently regular* $\mathbf{z}$*,*

$$\|R_{\Delta t}(\mathbf{z};\theta)\| \;=\; \Delta t \cdot \left\| \tfrac{1}{2}\big[\mathcal{A}_{\mathbf{D}},\, \mathcal{F}_\theta\big]\mathbf{z} \right\| \;+\; O(\Delta t^2), \tag{9}$$

where $\mathcal{F}_\theta \mathbf{z} := f_\theta(\mathbf{z})$ *and* $[\cdot,\cdot]$ *is the commutator. In particular* $R_{\Delta t} \to 0$ *as* $\Delta t \to 0$ uniformly in θ*, and* $R_{\Delta t}$ *does not depend on the observed data.*

Proof sketch. Expanding $S_{\Delta t}$ by BCH and subtracting the exact generator leaves the leading commutator term at order Δt; all data-dependent quantities cancel because $S_{\Delta t}$ and the generator are evaluated at the same state $\mathbf{z}$. □

Two consequences follow. First, minimizing $\|R_{\Delta t}\|^2$ cannot fit the model to observations, since the observations do not appear in (9); it drives θ toward operators whose diffusive and reactive parts commute, an inductive bias unrelated to the data. Second, the gradient signal it supplies is $O(\Delta t)$ and therefore vanishes in the very limit in which the discretization is most faithful. The empirical scaling is confirmed in Table 2 (observed order 1.00).

The constraint retained instead is the *steady-state* residual (3), which asks that the measured configuration be a fixed point of the learned operator. This does involve the data, through $\mathbf{z}_T$, and it constrains $\mathbf{D}_\theta$ and f_θ jointly: the set of pairs satisfying it is a proper subset of parameter space, characterized next.

E Non-identifiability from a single snapshot

The following records the central epistemic limitation of the setting: an operator fitted to one stationary field is determined only on a measure-zero subset of its own domain. The argument is elementary — a C^1 image of a two-dimensional domain cannot fill $\mathbb{R}^C$ once $C \ge 3$, so the stationarity constraint simply says nothing off that image — and we state it as a theorem for its consequence rather than for its difficulty. That consequence is sharp and is not, in our reading, generally acknowledged in this literature: it means every counterfactual an operator of this class produces is a statement about the architecture, not about the data, and it predicts in advance the null result of Section 5.6.

Theorem 11 (Latent-space non-identifiability)**.** *Let $\mathbf{z}^\star \in C^2(\Omega; \mathbb{R}^C)$ with $\Omega \subset \mathbb{R}^2$ open, and suppose $C \geq 3$. Fix any $\mathbf{D} \in (\mathcal{S}^2_{++})^C$ and define the admissible set*

$$\mathcal{F}(\mathbf{D}, \mathbf{z}^\star) := \Big\{ f \in C^1(\mathbb{R}^C; \mathbb{R}^C) \ : \ \mathcal{A}_\mathbf{D}\mathbf{z}^\star(x) + f\big(\mathbf{z}^\star(x)\big) = 0 \ \ \forall x \in \Omega \Big\}. \tag{10}$$

If $\mathcal{F}(\mathbf{D}, \mathbf{z}^\star) \neq \emptyset$ then it is an infinite-dimensional affine subset of $C^1(\mathbb{R}^C; \mathbb{R}^C)$. Specifically, for every $\varphi \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ vanishing on $\mathcal{R} := \mathbf{z}^\star(\Omega)$ and every $f \in \mathcal{F}$, one has $f + \varphi \in \mathcal{F}$, and the space of such φ is infinite-dimensional.

Proof. The constraint restricts f only through its values on $\mathcal{R} = \mathbf{z}^\star(\Omega)$; hence $f + \varphi$ satisfies it whenever $\varphi|_\mathcal{R} \equiv 0$. It remains to show that the space of such φ is infinite-dimensional. Since $\mathbf{z}^\star$ is C^1 on a 2-dimensional domain, $\mathcal{R}$ is a countable union of Lipschitz images of compact subsets of $\mathbb{R}^2$ and therefore has Hausdorff dimension at most 2. As $C \geq 3$, $\dim_H(\mathcal{R}) \leq 2 < C$, so $\mathcal{R}$ is Lebesgue-null in $\mathbb{R}^C$ and has empty interior. Its complement therefore contains an open ball B; choosing countably many disjoint balls $B_i \subset B$ and bump functions $\varphi_i \in C^\infty_c(B_i; \mathbb{R}^C)$ yields a linearly independent family, all vanishing on $\mathcal{R}$. ☐

Corollary 12 (What a stationarity fit can determine)**.** *Under the hypotheses of Theorem 11, the steady-state loss* (3) *constrains f_θ only on $\mathcal{R}$, a set of Lebesgue measure zero in $\mathbb{R}^C$. Any statement about f_θ off $\mathcal{R}$—in particular any counterfactual obtained by displacing the latent state away from the observed manifold—is determined by the inductive bias of the architecture and the regularizers, not by the data.*

Corollary 12 has a direct experimental consequence. The in-silico perturbation of Section 5.6 displaces $\mathbf{z}$ along a latent direction and integrates forward; by construction this evaluates f_θ at states off $\mathcal{R}$, where Theorem 11 shows the data exert no constraint. The null outcome reported there is therefore consistent with the theory rather than merely an empirical disappointment: a single stationary snapshot does not contain the information such a counterfactual would require. Breaking the degeneracy requires observations that visit a C-dimensional neighborhood of $\mathcal{R}$—dense temporal sampling, or interventional data.

F Linear stability and Turing analysis

Let $\mathbf{z}^\star \in \mathbb{R}^C$ be a spatially homogeneous equilibrium, so $f_\theta(\mathbf{z}^\star) = 0$, and write $\mathbf{J} := \nabla f_\theta(\mathbf{z}^\star)$. Linearising (5) about $\mathbf{z}^\star$ and expanding the perturbation in the basis (4) gives, for each $\mathbf{k}$, the finite-dimensional system

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\delta\hat{\mathbf{z}}(\mathbf{k}) \ = \ M(\mathbf{k})\,\delta\hat{\mathbf{z}}(\mathbf{k}), \qquad M(\mathbf{k}) \ := \ \mathbf{J} - \mathrm{diag}\big(q_1(\mathbf{k}), \ldots, q_C(\mathbf{k})\big), \tag{11}$$

with q_c as in (6). Define the growth rate $\mu(\mathbf{k}) := \max \mathrm{Re}\,\mathrm{spec}\, M(\mathbf{k})$ and its angular maximum $\mu^\star(k) := \max_{|\mathbf{k}|=k} \mu(\mathbf{k})$; the latter is required because anisotropy makes q_c direction-dependent.

Definition 13 (Turing instability)**.** The operator exhibits a *diffusion-driven (Turing) instability* at $\mathbf{z}^\star$ if $\mu^\star(0) < 0$ and $\mu^\star(k) > 0$ for some $k > 0$.

The first condition states that the reaction alone is stable; the second that diffusion destabilises a band of finite wavelengths. The selected wavelength is $\lambda^\star = 2\pi/k^\star$ with $k^\star = \arg\max_k \mu^\star(k)$, converted to microns by the stored coordinate scale.

Proposition 14 (Pure diffusion cannot pattern)**.** *If $f_\theta \equiv 0$ then $\mu^\star(k) = -\min_c \min_{|\mathbf{k}|=k} q_c(\mathbf{k}) \leq -\lambda_{\min}(\mathbf{D})k^2 \leq 0$, strictly decreasing in k. No Turing instability exists.*

Proof. $M(\mathbf{k}) = -\mathrm{diag}(q_c(\mathbf{k}))$ is diagonal with real non-positive entries; its largest eigenvalue is $-\min_c q_c(\mathbf{k})$, and $q_c(\mathbf{k}) \geq \lambda_{\min}|\mathbf{k}|^2$ by Lemma 2. □

Proposition 15 (Two-channel criterion)**.** *Let $C = 2$ with isotropic tensors $\mathbf{D}_c = d_c I$, $\mathbf{J} = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, and write $\sigma = |\mathbf{k}|^2$. Then $\mu(\mathbf{k}) > 0$ for some $\sigma > 0$ while $\mu(0) < 0$ if and only if*

$$a + d < 0, \qquad ad - bc > 0, \qquad d_2 a + d_1 d > 2\sqrt{d_1 d_2 (ad - bc)}. \tag{12}$$

Proof sketch. M has trace $a + d - (d_1 + d_2)\sigma$ and determinant $h(\sigma) := d_1 d_2 \sigma^2 - (d_2 a + d_1 d)\sigma + (ad - bc)$. The first two conditions give stability at $\sigma = 0$; the trace is then negative for all $\sigma > 0$, so instability requires $h(\sigma) < 0$ for some $\sigma > 0$. The minimum of the convex parabola h is attained at $\sigma^\star = (d_2 a + d_1 d)/(2 d_1 d_2)$ and is negative precisely under the third condition. □

Condition (12) requires the two channels to have *unequal* diffusivities of opposite-signed feedback—the classical short-range-activation / long-range-inhibition requirement. The operators fitted in Section 5 satisfy the first two conditions but not the third: the dispersion curve is monotonically decreasing (Fig. 8c), and $\mu^\star(k) < 0$ for all $k > 0$ in every run.

Remark 16 (Why the negative result is expected)*.* The training objective (3) rewards making the *observed* configuration stationary. A Turing-unstable operator does the opposite: it renders the homogeneous state unstable so that structure grows spontaneously. Since the observed field is treated as a fixed point rather than as the outcome of an instability, the fitted operator is driven toward dissipativity. Detecting pattern-forming instability would require observations away from steady state, consistent with Corollary 12.

G Discretization invariance of the encoder

The encoder must map measurements on an arbitrary point set to a field. Let μ be the sampling measure on Ω and consider the kernel-integral operator

$$(\mathcal{K}v)(x) \;=\; \int_\Omega \kappa_\theta(y - x) \odot v(y)\, \mathrm{d}\mu(y), \tag{13}$$

of which the graph layer is the empirical counterpart $(\mathcal{K}_N v)(x_i) = |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(x_j - x_i) \odot v(x_j)$.

Proposition 17 (Monte-Carlo consistency)**.** *If $\{x_j\}_{j=1}^N$ are i.i.d. draws from μ and $\kappa_\theta \odot v \in L^2(\mu)$, then for each x, $\mathbb{E}\big[(\mathcal{K}_N v)(x)\big] = (\mathcal{K}v)(x)$ and $\mathrm{Var}\big[(\mathcal{K}_N v)(x)\big] = O(N^{-1})$; consequently $\|\mathcal{K}_N v - \mathcal{K}v\|_{L^2(\mu)} = O_P(N^{-1/2})$.*

Proof. The empirical operator is a sample mean of i.i.d. terms with finite second moment; unbiasedness and the $O(N^{-1})$ variance are immediate, and the norm bound follows by Chebyshev and Fubini. □

Proposition 17 is the sense in which the encoder is discretization-invariant: the population operator (13) depends on the geometry through $\kappa_\theta(y - x)$ alone and not on the sampling pattern, and the empirical operator converges to it as the sampling density increases, whatever the lattice. **This is an asymptotic statement about the encoder, and it is worth emphasising that it does not imply transfer of the fitted operator.** The transfer experiments of Section 5 show empirically that it does not: $\mathbf{D}_\theta$ and f_θ are fitted to one tissue's stationary field, and Theorem 11 indicates they are unconstrained away from that field's range. Discretization invariance of the *architecture* and portability of the *fit* are distinct properties, and only the former is established here.

Proposition 18 (Splatting consistency)**.** *Let $w_\sigma(r) = e^{-r^2/2\sigma^2}$ and define the Nadaraya–Watson field $F_\sigma(u) = \sum_i w_\sigma(\|u - x_i\|) v_i / \sum_i w_\sigma(\|u - x_i\|)$. If $v_i = v(x_i)$ for $v \in C^2$ and the sampling density p is bounded away from zero, then $F_\sigma(u) = v(u) + O(\sigma^2) + O_P((N\sigma^2)^{-1/2})$.*

Proof sketch. Standard kernel-regression bias–variance decomposition in two dimensions: the bias term is $\frac{\sigma^2}{2}\left(\Delta v + 2\nabla v \cdot \nabla \log p\right) + o(\sigma^2)$ and the variance is $O((N\sigma^2)^{-1})$. $\square$

The learnable bandwidth σ therefore trades a smoothing bias against sampling noise. Since the bias term is proportional to Δv, splatting contributes a further diffusive smoothing on top of the operator—a mechanism that plausibly compounds the over-smoothing reported in Section 5, and one that the present work does not disentangle.

Complexity

Let N be the number of measured locations, G genes, C latent channels, $H \times W$ the lattice, E the number of graph edges and T the integration horizon. The costs per forward pass are: gene projection $O(NGd)$; graph layers $O(LEd)$; splatting $O(Nr^2C)$ for radius r; spectral blocks $O(BCHW \log HW)$; operator $O(TCHW \log HW)$; decoding $O(NCd + NGd)$. The operator term is therefore linear in the horizon and quasi-linear in the lattice, and for the configurations used here (C=32, H=W=64, T=8) it is under 4% of the forward cost—consistent with the parameter count of Section 6, where the operator holds 0.43% of the weights yet accounts for the entire margin over the strongest baseline.

H Empirical details and full result tables

This appendix collects the complete result tables underlying Section 5. Tables are placed inline with `[h]` so that each appears with its heading.

H.1 Single-section model comparison

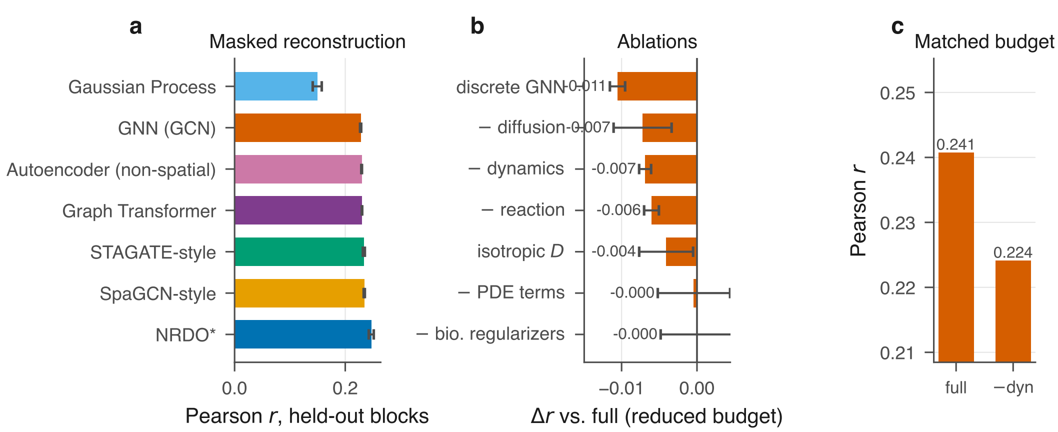

Figure 3: **Locating the effect** on the primary section, where all nine models are run. (a) Full model comparison. (b) Ablation deltas at the reduced budget. (c) The two decisive controls at the benchmark budget: disabling the operator (T=0) costs more than the full margin over the best baseline.

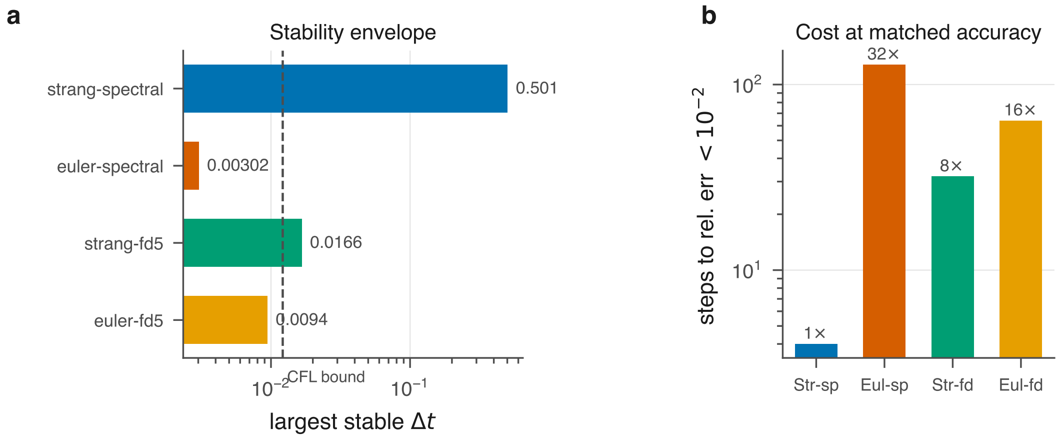

Figure 4: **What the exact spectral solve buys.** (a) Largest stable step size per scheme against the CFL bound for the explicit five-point Laplacian. (b) Steps required to integrate a fixed horizon to relative error $< 10^{-2}$, relative to the exponential scheme.

Table 3: Paired comparison of NRDO against each baseline, with the tissue section as the unit of analysis. Δ is the mean paired difference in Pearson r (positive favors NRDO) with a percentile bootstrap 95% CI over sections; d_z is Cohen's effect size for paired designs; p is the Wilcoxon signed-rank test and p_{Holm} its Holm–Bonferroni correction across the baseline family.

baseline	n	Δr	95% CI	d_z	p	p_{Holm}	wins
SpaGE-style	17	+0.1416	[+0.1217, +0.1606]	3.36	1.5e-05	4.6e-05	17/17
Neural field (SIREN)	23	+0.0635	[+0.0514, +0.0765]	2.02	2.4e-07	1.7e-06	23/23
GP, multi-bandwidth	23	+0.0483	[+0.0402, +0.0569]	2.32	2.4e-07	1.7e-06	23/23
Tangram-style	23	+0.0194	[+0.0137, +0.0250]	1.37	4.5e-06	1.8e-05	21/23
GNN (GCN)	23	+0.0146	[+0.0097, +0.0199]	1.14	2.4e-06	1.2e-05	22/23
Autoencoder (non-spatial)	23	+0.0135	[+0.0077, +0.0197]	0.89	0.00021	0.00043	18/23
STAGATE-style	23	+0.0117	[+0.0058, +0.0180]	0.75	0.0027	0.0027	17/23

Table 4: Numerical behavior of the integrator variants. The largest stable step is measured by sweeping Δt over $[10^{-4}, 0.501]$ and testing boundedness over 300 steps; the CFL reference is $h^2/(4\lambda_{\max}) = 0.01215$ for the explicit five-point Laplacian. **strang-spectral never destabilized within the sweep** ($\Delta t \leq 0.501$), so its entry is a lower bound set by the grid rather than a measured threshold, consistent with the unconditional stability of Proposition 5. Cost is the number of steps required to integrate a fixed horizon to relative error $< 10^{-2}$, relative to the exponential scheme; the achieved error is reported alongside, since the step grid is coarse and the schemes do not land on the same accuracy. The empirical limit for `euler-fd5` tracks the CFL bound, confirming the analysis.

scheme	largest stable Δt	vs. CFL	steps to 10^{-2}	achieved err.	cost
strang-spectral	≥0.501	≥41×	4	8.4e-03	1×
euler-spectral	0.00302	0.25×	128	2.3e-03	32×
strang-fd5	0.0166	1.4×	32	9.1e-04	8×
euler-fd5	0.0094	0.77×	64	6.1e-03	16×

Table 5: Biological preservation of the reconstruction, averaged over 10 sections with reference annotations (Space Ranger clusters, Xenium clusters, and curated anatomical regions). Predicted expression at held-out locations is clustered and scored against the reference; ARI retention is the ratio to the score obtained from the measured data, so 1.0 would mean the reconstruction is as informative as the measurement. Marker AUROC asks whether the predicted field still discriminates a region using that region's measured markers; k-NN preservation is the overlap of neighborhoods in measured versus predicted expression space; $|\Delta C|$ is the error in Geary's C.

model	ARI ↑	ARI ret. ↑	NMI ↑	marker AUROC ↑	k-NN pres. ↑	$\lvert\Delta C\rvert$ ↓
GNN (GCN)	0.206 ± 0.147	-0.256 ± 2.333	0.349 ± 0.164	0.781 ± 0.055	0.157 ± 0.073	0.754 ± 0.061
STAGATE-style	0.205 ± 0.138	-0.135 ± 1.902	0.348 ± 0.157	0.787 ± 0.055	0.158 ± 0.074	0.739 ± 0.059
GP, multi-bandwidth	0.203 ± 0.135	-0.316 ± 2.243	0.361 ± 0.167	0.786 ± 0.091	0.187 ± 0.086	0.741 ± 0.071
Neural field (SIREN)	0.085 ± 0.055	0.151 ± 0.087	0.169 ± 0.104	0.698 ± 0.055	0.098 ± 0.053	**0.534** ± **0.117**
NRDO (ours)	**0.224** ± **0.109**	**1.127** ± **2.040**	**0.373** ± **0.160**	**0.797** ± **0.059**	**0.188** ± **0.094**	0.794 ± 0.064

Table 6: Processed spatial transcriptomics datasets. Locations and genes are counts after QC and gene selection; extent is the physical bounding box of the section.

dataset	technology	organism	locations	genes	extent (μm)	resolution
merfish_allen_01	MERFISH, 500-gene pane	Mus	23,376	500	4178×4097	single cell
merfish_allen_05	MERFISH, 500-gene pane	Mus	45,896	501	6127×5166	single cell
merfish_allen_10	MERFISH, 500-gene pane	Mus	42,070	503	8402×5441	single cell
merfish_allen_14	MERFISH, 500-gene pane	Mus	94,357	500	9048×5717	single cell
merfish_allen_18	MERFISH, 500-gene pane	Mus	44,309	500	7980×4825	single cell
merfish_allen_26	MERFISH, 500-gene pane	Mus	79,282	501	9460×5685	single cell
merfish_allen_30	MERFISH, 500-gene pane	Mus	91,017	501	9870×5843	single cell
merfish_allen_31	MERFISH, 500-gene pane	Mus	91,951	502	9837×6094	single cell
merfish_allen_36	MERFISH, 500-gene pane	Mus	115,675	503	10074×6559	single cell
merfish_allen_40	MERFISH, 500-gene pane	Mus	104,461	502	10000×6961	single cell
merfish_allen_42	MERFISH, 500-gene pane	Mus	75,433	503	9875×7086	single cell
merfish_allen_50	MERFISH, 500-gene pane	Mus	92,787	501	8806×6604	single cell
mosta_embryo_E10.5	Stereo-seq, DNA nanoba	Mus	18,360	2,099	3500×4950	bin50 (25 um)
mosta_embryo_E9.5	Stereo-seq, DNA nanoba	Mus	5,898	2,069	1975×2650	bin50 (25 um)
perturb_norman	Perturb-seq (CRISPRa),	Homo	111,659	2,073	–	single cell
visium_ffpe_human_prostate	Visium FFPE (55 um spo	Homo	4,369	2,088	6363×6706	spot (multi-cell)
visium_ffpe_mouse_brain	Visium FFPE (55 um spo	Mus	2,264	2,075	4959×5307	spot (multi-cell)
visium_human_breast	Visium (55 um spots, h	Homo	4,869	2,087	6395×6750	spot (multi-cell)
visium_human_heart	Visium (55 um spots, 1	Homo	331	2,051	5877×6040	spot (multi-cell)
visium_human_lymph_node	Visium (55 um spots, 1	Homo	4,025	2,073	5967×6744	spot (multi-cell)
visium_mouse_brain	Visium (55 um spots, 1	Mus	2,691	2,079	4822×6379	spot (multi-cell)
visium_mouse_brain_coronal	Visium (55 um spots, 1	Mus	2,896	2,082	4510×6689	spot (multi-cell)
visium_mouse_kidney	Visium (55 um spots, 1	Mus	1,433	2,079	4974×4894	spot (multi-cell)
xenium_mouse_brain	Xenium In Situ, 248-ge	Mus	36,362	248	5212×3128	single cell

Table 7: Masked spatial reconstruction across 23 tissue sections spanning four technologies; mean ± s.d. over sections, 2 seeds each. Excluded for incomplete coverage: Gaussian Process (1/23), SpaGE-style (17/23). Stars: Holm-corrected Wilcoxon signed-rank p against NRDO, section as the unit of analysis ($^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$). Win record and effect sizes: Table 3.

model	n	Pearson r ↑	SSIM ↑	$\lvert\Delta I\rvert$ ↓
STAGATE-style	23	0.165 ± 0.067**	0.319 ± 0.064*	0.764 ± 0.049***
Autoencoder (non-spatial)	23	0.163 ± 0.067***	0.256 ± 0.083***	0.755 ± 0.054***
GNN (GCN)	23	0.162 ± 0.065***	0.317 ± 0.063*	0.791 ± 0.047***
Tangram-style	23	0.157 ± 0.068***	**0.338** ± **0.076**	0.794 ± 0.073***
GP, multi-bandwidth	23	0.129 ± 0.056***	0.295 ± 0.055***	0.759 ± 0.064***
Neural field (SIREN)	23	0.113 ± 0.054***	0.251 ± 0.056***	**0.593** ± **0.111*****
NRDO (ours)	23	**0.177** ± **0.068**	0.329 ± 0.068	0.844 ± 0.048

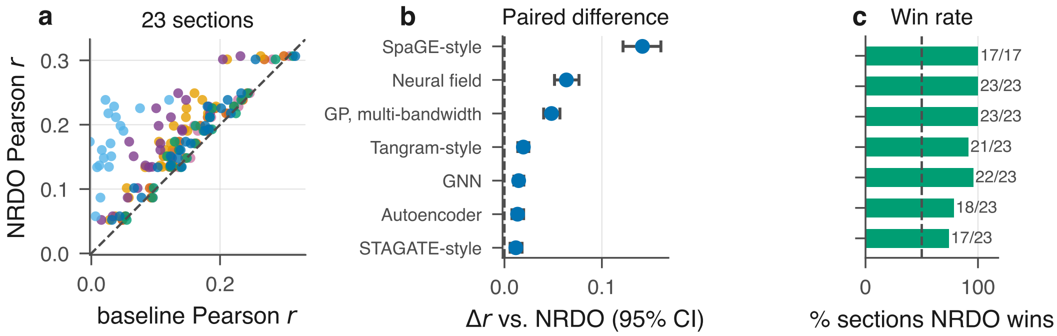

Figure 5: **Benchmark across 22 tissue sections.** (a) NRDO against each baseline, one point per section. (b) Mean paired difference with a percentile bootstrap 95% CI. (c) Fraction of sections on which NRDO wins. Full table and paired statistics in Appendix H.

Table 8: Held-out Pearson r under input corruption. Each model is re-trained under each corruption with identical splits, since the realistic deployment is training on the data one has. Columns are increasing corruption level (for *density*, decreasing fraction of the section retained).

model	L1	L2	L3	L4
noise				
STAGATE-style	0.208	0.195	0.177	0.124
NRDO (ours)	0.201	0.187	0.170	0.118
GNN (GCN)	0.198	0.185	0.168	0.115
dropout				
STAGATE-style	0.208	0.207	0.206	0.206
NRDO (ours)	0.201	0.196	0.197	0.193
GNN (GCN)	0.198	0.191	0.194	0.188
density				
NRDO (ours)	0.195	0.158	0.197	0.201
STAGATE-style	0.173	0.163	0.187	0.208
GNN (GCN)	0.169	0.149	0.169	0.198
knn				
STAGATE-style	0.208	0.208	0.209	0.209
NRDO (ours)	0.201	0.201	0.201	0.201
GNN (GCN)	0.199	0.198	0.197	0.195

H.2 Numerical behavior

H.3 Paired statistics, numerics and biology

H.4 Dataset inventory

H.5 Multi-section benchmark

H.6 Robustness

H.7 Spectral matching

Full discussion

Section 5 reports that our predicted fields are too smooth, and Section H.8 shows this is structural rather than an artifact of the training budget. The natural remedy is a term matching the radially averaged power spectrum of the predicted field to that of the measured field. We implement it, sweep its weight, and report the trade-off (Table 9, Fig. 6; 2 seeds per point).

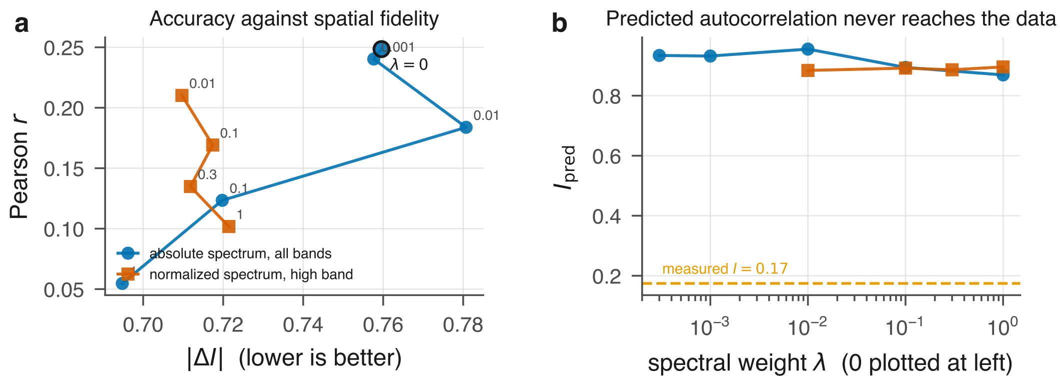

Figure 6: **Spectral matching.** (a) The trade-off: up and to the left is better. Points are labeled by weight λ; $\lambda = 0$ is the model used everywhere else. (b) Predicted global autocorrelation against λ. No weight of either form brings I_{pred} near the measured value.

The form of the term matters more than its weight. Matching the *absolute* spectrum over all bands leaves an amplitude degree of freedom the model can exploit without redistributing energy across scales — a controlled check finds it penalizes a purely rescaled field more heavily than a genuinely smoothed one — and its best setting costs -78% of the Pearson r to gain -9% of $|\Delta I|$. Matching the spectrum *normalized* by total power over the upper half of the band removes that freedom and penalizes the frequencies a dissipative operator actually loses. At $\lambda = 0.01$ it reaches $|\Delta I| = 0.710$, better than the absolute form achieves at any weight, for -15% of the accuracy; the curve turns over above that, so it is an optimum rather than a point on a slope.

The conclusion is a bound, not a fix. Across every weight and both forms the predicted autocorrelation bottoms out at $I_{\text{pred}} = 0.869$ against a measured 0.174: weighting the term heavily enough to destroy predictive accuracy still does not bring the reconstruction near the roughness of the data. We keep $\lambda = 0$ elsewhere and conclude that the smoothness belongs to the operator class rather than to the objective.

Table 9: Spectral matching: the trade-off between reconstruction accuracy and spatial fidelity as the term's weight λ is swept, 2 seeds per point, on the primary section. $\lambda = 0$ is the model used everywhere else in this paper. The measured field has $I = 0.174$, so a faithful reconstruction would drive I_{pred} down toward that value. Two forms are compared: matching the absolute power spectrum across all radial bands, and matching the spectrum normalized by total power over the upper half of the band. The first leaves an amplitude degree of freedom that the model can exploit without redistributing energy across scales; the second removes it and penalizes only the frequencies a dissipative operator actually loses.

weight	Pearson r ↑	SSIM ↑	$\lvert\Delta I\rvert$ ↓	I_{pred}	$\lvert\Delta C\rvert$ ↓
absolute spectrum, all bands					
$\lambda = 0$	0.249 ± 0.005	0.405 ± 0.007	0.760 ± 0.004	0.934 ± 0.004	0.727 ± 0.003
$\lambda = 0.001$	0.240 ± 0.004	0.399 ± 0.004	0.758 ± 0.003	0.932 ± 0.003	0.728 ± 0.002
$\lambda = 0.01$	0.184 ± 0.007	0.368 ± 0.017	0.781 ± 0.018	0.955 ± 0.018	0.745 ± 0.010
$\lambda = 0.1$	0.123 ± 0.004	0.405 ± 0.002	0.720 ± 0.001	0.894 ± 0.000	0.711 ± 0.000
$\lambda = 1$	0.055 ± 0.007	0.401 ± 0.023	0.695 ± 0.003	0.869 ± 0.003	0.698 ± 0.003
normalized spectrum, high band					
$\lambda = 0.01$	0.210 ± 0.004	0.384 ± 0.001	0.710 ± 0.003	0.884 ± 0.003	0.702 ± 0.001
$\lambda = 0.1$	0.169 ± 0.000	0.274 ± 0.000	0.717 ± 0.002	0.892 ± 0.002	0.707 ± 0.000
$\lambda = 0.3$	0.135 ± 0.003	0.266 ± 0.001	0.712 ± 0.000	0.886 ± 0.000	0.703 ± 0.000
$\lambda = 1$	0.102 ± 0.000	0.256 ± 0.002	0.721 ± 0.002	0.896 ± 0.002	0.709 ± 0.002

H.8 Converged single-section comparison

Full discussion

The benchmark trains every model for a fixed, reduced number of epochs, so its ordering could be a fact about the budget rather than the models. We settle this on visium_mouse_brain, the section where the ordering changes with budget, training every model to a shared early-stopping criterion on the full section with 3 seeds (Appendix H, Table 10). NRDO attains 0.245 ± 0.006 against 0.232 ± 0.001 for the strongest baseline (STAGATE-style), winning on 3/3 seeds against every comparator with $d_z = 1.99$–2.84, having *lost* to that baseline at the benchmark budget. The budget is therefore conservative for NRDO here rather than neutral: Table 7 establishes consistency across anatomy at a fixed budget, not that its margins are the converged margins.

Two findings survive convergence and matter more than the ordering. The non-spatial autoencoder still matches the graph models (the best leads it by 0.004 Pearson r), so their near-tie is a property of this task rather than of undertraining. And NRDO retains the largest error in Moran's I (0.760 against 0.705), so the over-smoothing is structural. NRDO costs 653 s against 100–181 s per run on one CPU.

Table 10: Converged comparison on visium_mouse_brain: the full section, no subsampling, trained to a shared early-stopping criterion on held-out validation rather than to a fixed epoch count, 3 seeds. d_z and the win record are paired over seeds against NRDO. This addresses whether the reduced-budget ordering of Table 7 is the converged ordering. It is not: at 200 epochs on this section NRDO scores below the STAGATE-style baseline, and at convergence it scores above it, so the benchmark budget is conservative for NRDO here rather than neutral. Two further points survive convergence — the non-spatial autoencoder still matches the graph models, so their near-tie is a property of the task and not of undertraining; and NRDO retains the largest error in Moran's I, so the over-smoothing is structural. Wall time is single-CPU and is reported because NRDO is several times more expensive than every baseline.

model	Pearson r ↑	RMSE ↓	SSIM ↑	$\lvert\Delta I\rvert$ ↓	$\lvert\Delta C\rvert$ ↓	wall	d_z	wins
STAGATE-style	0.232 ± 0.001	0.255 ± 0.001	0.393 ± 0.002	0.708 ± 0.002	0.683 ± 0.001	181 s	1.99	3/3
Autoencoder (non-spatial)	0.228 ± 0.002	0.257 ± 0.001	0.392 ± 0.003	0.705 ± 0.004	0.680 ± 0.003	100 s	2.25	3/3
GNN (GCN)	0.227 ± 0.001	0.255 ± 0.001	0.391 ± 0.003	0.724 ± 0.008	0.692 ± 0.006	170 s	2.84	3/3
NRDO (ours)	$\mathbf{0.245} \pm 0.006$	$\mathbf{0.256} \pm 0.007$	$\mathbf{0.399} \pm 0.009$	$\mathbf{0.760} \pm 0.009$	$\mathbf{0.726} \pm 0.007$	653 s	–	–

H.9 Null controls for the recovered diffusion length

Table 11: Null controls for the recovered diffusion length. Every row refits the identical architecture at the identical budget on the primary section. Shuffling coordinates destroys all spatial structure while preserving every gene's marginal; those rows reach $r \approx 0$ yet recover the length scale of the *untrained* model, so the quantity is close to its initialization rather than inferred from tissue. The bandwidth and lattice sweeps show why neither is the mechanism: the length is invariant in microns across a fourfold change in pitch while the same quantity in lattice cells moves by the same factor.

condition	length (μm)	length (cells)	held-out r
measured tissue, H=64	374.6 ± 1.7	3.76	0.248
splat $\sigma = 0.5$ cells	375.5 ± 0.9	3.77	0.247
splat $\sigma = 2$ cells	377.7 ± 1.4	3.79	0.247
splat $\sigma = 4$ cells	376.4 ± 0.7	3.77	0.247
lattice H=32 (pitch 199 μm)	376.3 ± 0.2	1.89	0.245
lattice H=128 (pitch 50 μm)	378.2 ± 5.6	7.58	0.243
coordinates shuffled, H=64	404.5 ± 0.1	4.06	0.004
coordinates shuffled, H=128	404.7 ± 0.1	8.12	0.004

The rows above the rule refit the model on measured tissue under varied encoder and discretization settings; the rows below refit it on sections whose expression has been permuted across positions.

The shuffled fits attain $r \approx 0$, as they must, and yet return a length scale within -0.06% of the untrained model's 405 μm. Measured tissue moves the same quantity by -7.4%. Section 5 reports what this does and does not license.

H.10 Ablation sweep

Full sweep at the reduced budget; the two matched-budget controls are reported in Section 6.

Table 12: Ablations on the mouse-brain Visium section. Each variant removes exactly one ingredient with all else held fixed. Δr is the change in held-out Pearson r relative to the full model. The $-$ *dynamics* row is the critical control: it disables the operator entirely (T=0) while leaving its parameters allocated, so the two rows are exactly parameter-matched. **This table uses a smaller training budget than the main benchmark** (fewer epochs and seeds); it is internally consistent, but absolute values are not comparable across the two tables.

variant	params	Pearson $r \uparrow$	Δr
Full model	2.20M	**0.233** $\pm$ **0.005**	–
$-$ dynamics (T=0)	2.20M	0.227 $\pm$ 0.001	-0.007
$-$ diffusion term	2.20M	0.226 $\pm$ 0.004	-0.007
$-$ reaction term	2.19M	0.227 $\pm$ 0.001	-0.006
$-$ PDE constraints	2.20M	0.233 $\pm$ 0.005	-0.000
$-$ biological regularizers	2.20M	0.233 $\pm$ 0.005	-0.000
isotropic D	2.20M	0.229 $\pm$ 0.004	-0.004
state-dependent D	2.20M	0.236 $\pm$ 0.004	+0.003
discrete GNN operator	0.61M	0.223 $\pm$ 0.001	-0.011
latent C=8	1.07M	0.213 $\pm$ 0.005	-0.021
latent C=16	1.30M	0.220 $\pm$ 0.002	-0.013
latent C=64	5.77M	0.237 $\pm$ 0.000	+0.004

H.11 Cross-tissue and cross-resolution transfer

H.12 Developmental forecasting

Horizon sweep for E9.5 $\rightarrow$ E10.5. Consecutive MOSTA stages are distinct embryos, so the comparison is at the level of the rasterized field after isotropic normalization to a common frame.

H.13 In-silico morphogen source

Split-half protocol of Section 5.6: the perturbation direction is defined from a random half of each pathway and enrichment scored on the held-out half, against a permutation null over matched gene sets.

H.14 Perturb-seq consistency

An out-of-context probe: Norman et al. profiled non-spatial K562 cells whereas the operator is fitted to tissue. With $n = 7$ testable perturbations the comparison is underpowered and we report it as inconclusive.

H.15 Latent fields

I Reproducibility

All datasets download automatically with a recorded SHA256 manifest; every table and figure in this paper is regenerated from run artifacts by a single command; configurations, seeds and checkpoints are released.

Table 13: Cross-tissue, cross-species transfer: adult mouse brain → human breast carcinoma. Shared gene vocabulary: 390 genes. Mean ± s.d. over seeds.

model	Pearson r ↑	RMSE ↓	SSIM ↑
source (in-domain)			
NRDO (ours)	0.214 ± 0.004	0.252 ± 0.004	0.369 ± 0.006
STAGATE-style	0.214 ± 0.000	0.249 ± 0.000	0.362 ± 0.001
GNN (GCN)	0.206 ± 0.001	0.250 ± 0.000	0.363 ± 0.004
Gaussian Process	0.146 ± 0.002	0.291 ± 0.004	0.289 ± 0.003
target (zero-shot)			
STAGATE-style	0.071 ± 0.006	0.403 ± 0.001	0.285 ± 0.008
Gaussian Process	0.070 ± 0.004	0.453 ± 0.001	0.311 ± 0.008
GNN (GCN)	0.056 ± 0.009	0.415 ± 0.006	0.250 ± 0.004
NRDO (ours)	-0.005 ± 0.010	0.504 ± 0.016	0.165 ± 0.009
target (decoder-only fine-tune)			
STAGATE-style	0.129 ± 0.009	0.386 ± 0.001	0.387 ± 0.005
GNN (GCN)	0.112 ± 0.005	0.390 ± 0.003	0.378 ± 0.006
NRDO (ours)	0.067 ± 0.021	0.398 ± 0.005	0.300 ± 0.023
Gaussian Process	–	–	–
target (in-domain oracle)			
STAGATE-style	0.149 ± 0.000	0.375 ± 0.000	0.433 ± 0.003
GNN (GCN)	0.145 ± 0.001	0.377 ± 0.000	0.426 ± 0.001
NRDO (ours)	0.136 ± 0.002	0.397 ± 0.002	0.329 ± 0.003
Gaussian Process	0.068 ± 0.002	0.459 ± 0.001	0.301 ± 0.008
target (training-mean floor)			
mean	-0.000	0.408	0.342

Table 14: Cross-resolution transfer: 55 μm Visium spots → single-cell Xenium. Shared gene vocabulary: 248 genes. Mean ± s.d. over seeds.

model	Pearson r ↑	RMSE ↓	SSIM ↑
source (in-domain)			
NRDO (ours)	0.261 ± 0.006	0.246 ± 0.001	0.392 ± 0.005
STAGATE-style	0.246 ± 0.000	0.247 ± 0.001	0.370 ± 0.002
GNN (GCN)	0.236 ± 0.002	0.249 ± 0.001	0.363 ± 0.002
Gaussian Process	0.158 ± 0.005	0.291 ± 0.004	0.278 ± 0.011
target (zero-shot)			
STAGATE-style	0.242 ± 0.002	1.780 ± 0.002	0.631 ± 0.003
GNN (GCN)	0.234 ± 0.002	1.798 ± 0.005	0.618 ± 0.002
Gaussian Process	0.183 ± 0.008	2.114 ± 0.032	0.577 ± 0.003
NRDO (ours)	0.124 ± 0.017	1.855 ± 0.030	0.544 ± 0.016
target (in-domain oracle)			
NRDO (ours)	0.277 ± 0.005	1.745 ± 0.006	0.667 ± 0.004
STAGATE-style	0.269 ± 0.001	1.756 ± 0.002	0.677 ± 0.002
GNN (GCN)	0.267 ± 0.001	1.760 ± 0.001	0.675 ± 0.001
Gaussian Process	0.165 ± 0.003	2.142 ± 0.022	0.559 ± 0.017
target (training-mean floor)			
mean	-0.000	1.877	0.398

Table 15: Developmental forecasting on Stereo-seq: the E9.5 field is encoded, integrated forward for T operator steps, and scored against the measured E10.5 field. **Consecutive stages are different embryos**, so there is no cell-to-cell correspondence and the comparison is made at the level of the rasterized field after isotropic normalization to a common frame; it inherits the error of that coarse registration. *Persistence* predicts E10.5 = E9.5 and is the reference the operator must beat for its dynamics to carry temporal information.

model	horizon	field $r \uparrow$	field RMSE $\downarrow$	genes with $r > 0$
Persistence (no dynamics)	–	0.064	0.346	61%
Spatial-mean predictor	–	0.000	0.257	4%
NRDO	T=0	0.082 $\pm$ 0.001	0.320	74%
NRDO	T=2	0.089 $\pm$ 0.000	0.305	72%
NRDO	T=4	0.092 $\pm$ 0.000	0.297	73%
NRDO	T=8	0.094 $\pm$ 0.000	0.289	72%
NRDO	T=16	0.095 $\pm$ 0.001	0.283	72%
NRDO	T=32	0.090 $\pm$ 0.003	0.285	70%

Table 16: In-silico morphogen source (*bead implant*). The latent perturbation direction is defined from a random half of each pathway's genes; the table reports where the *held-out* half ranks in the response magnitude distribution (lower is stronger; 50% is chance). p is a permutation test over random gene sets of matched size. Response half-width is the distance at which the predicted response falls to half its peak. $^*p < 0.05$.

pathway	genes	held-out rank (%)	null (%)	p	half-width (μm)
BMP	22	48.6 $\pm$ 6.0	49.7	0.475	575 $\pm$ 154
FGF	19	72.6 $\pm$ 0.8	50.3	0.992	618 $\pm$ 58
SHH	15	56.6 $\pm$ 6.7	50.4	0.698	466
WNT	26	56.4 $\pm$ 3.9	49.9	0.762	649 $\pm$ 143

Table 17: Agreement between the model's counterfactual gene–gene responses and measured CRISPRa responses from Norman et al. (2019). **This is an out-of-context probe**: the Perturb-seq data are non-spatial K562 cells while the operator is fitted to human breast tissue, so it tests whether the reaction module encodes generic transcriptional coupling, not whether it recovered tissue-specific signaling. Null is a within-perturbation label permutation; p is a Wilcoxon signed-rank test against that null.

model	#perturb.	Spearman ρ	null	% positive	p
STAGATE-style	7	0.010 $\pm$ 0.000	+0.004	64%	0.633
NRDO (ours)	7	-0.018 $\pm$ 0.009	-0.009	57%	0.844

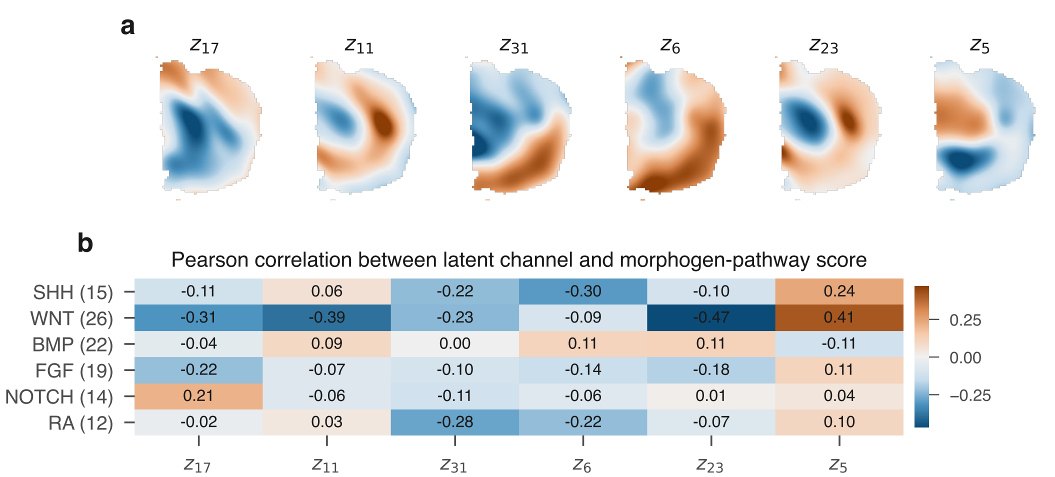

Figure 7: Latent channels of the relaxed field (a) and their correlation with canonical morphogen-pathway scores (b). Section 5.6 shows these correlations do not survive a causal split-half test.

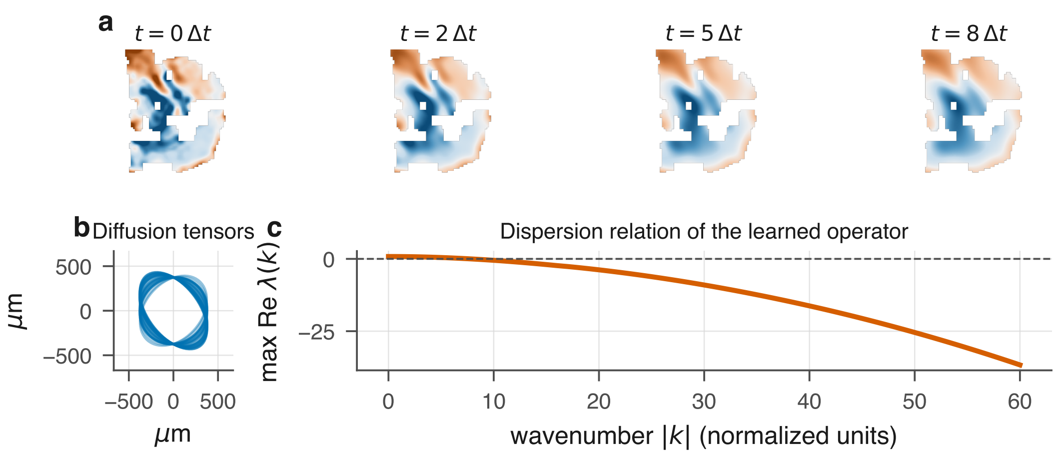

Figure 8: **The fitted operator.** (a) A latent channel relaxing under the learned dynamics. (b) Per-channel diffusion tensors as length-scale ellipses, showing the anisotropy is used. (c) Dispersion relation of the linearized operator, which decreases monotonically in $|\mathbf{k}|$ and so supports no Turing band.